AGI

Benchmarks
Feasibility
Implications

Frederik Schittny | Institute of Theoretical Informatics, KIT
16.02.2026


Sam Altman
@sama
In from three to eight years we will have a machine with the general intelligence of an average human being
8:11 PM · Mar 31, 2025 · 5.5M Views
2K
4K
58K
3.4K

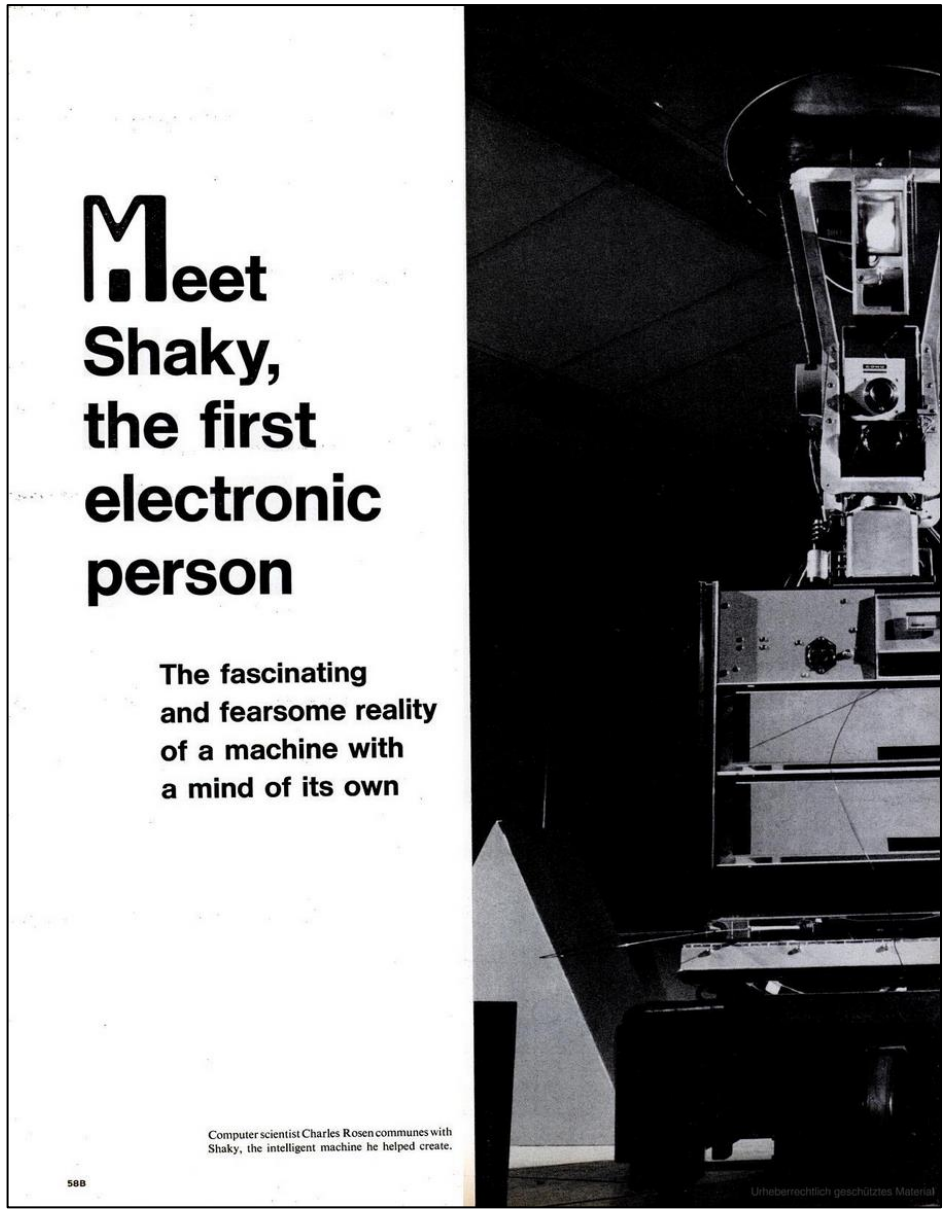

Marvin Minsky
@marv_minsky
In from three to eight years we will have a machine with the general intelligence of an average human being
8:11 PM · Mar 31, 2025 · 5.5M Views
2K
4K
58K
3.4K



"In from three to eight years we will have a machine with the general intelligence of an average human being."

– Marvin Minsky, *Life* Magazine, November 1970

[0][Life Magazine, 1970]



"In from three to eight years we will have a machine with the general intelligence of an average human being."

– Marvin Minsky, *Life* Magazine, November 1970

[0] [Life Magazine, 1970]

Structure

- AGI Definition – *Ambiguous Foundations*
- Benchmarks – *Some Proposals*
- Feasibility – *Timelines and Approaches*
- Implications – *Propagating Inaccuracy*
- Discussion



[1][Russel & Norvig, 2020] [2][Hendrycks et al., 2025] [3][Hutson, 2025]

Defining Intelligence (is really hard)

emotions

philosophy

abstraction

internal processes

behavior

reasoning

mathematics

psychology

neuroscience

creativity

rationality

“human”

biology

learning

sociology

self-awareness

[2][Hendrycks et al., 2025] [3][Hutson, 2025]

The most important question to keep in mind when talking about AGI:

What is the underlying definition of AGI in this context?

The Turing Test

- Indistinguishability in conversation[4]
  - Beat to various degrees
  - Language models “specialized” for this test
- Criticism
  - Can be “faked” easily[5]
  - Language bad indicator for intelligence[6, 7]

Underlying intelligence understanding:

Indistinguishable (language) between human and AI

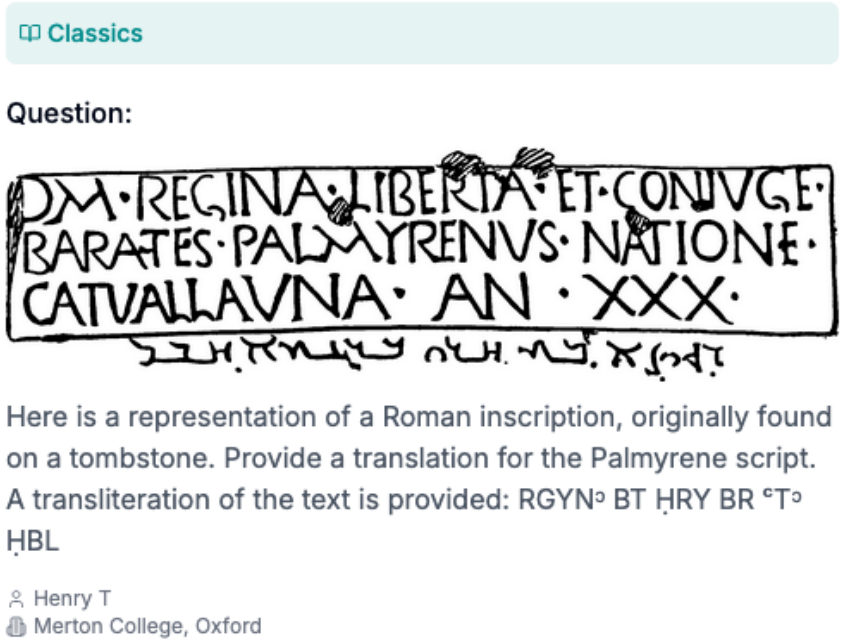


[xkcd 329]

[4][Turing, 1949] [5][Weizenbaum, 1966] [6][Fedorenko et al., 2025] [7][The Verge, 2025]

Humanity's Last Exam

- Intended for foundation models
- Traditional LLM benchmarks "too easy"
  - Solution: make the questions harder
- 2500 expert level questions
  - Mostly scientific
  - "Multimodal" (text and images)



[lastexam.ai]

Underlying intelligence understanding:

Expert level knowledge and reasoning skills across many topics

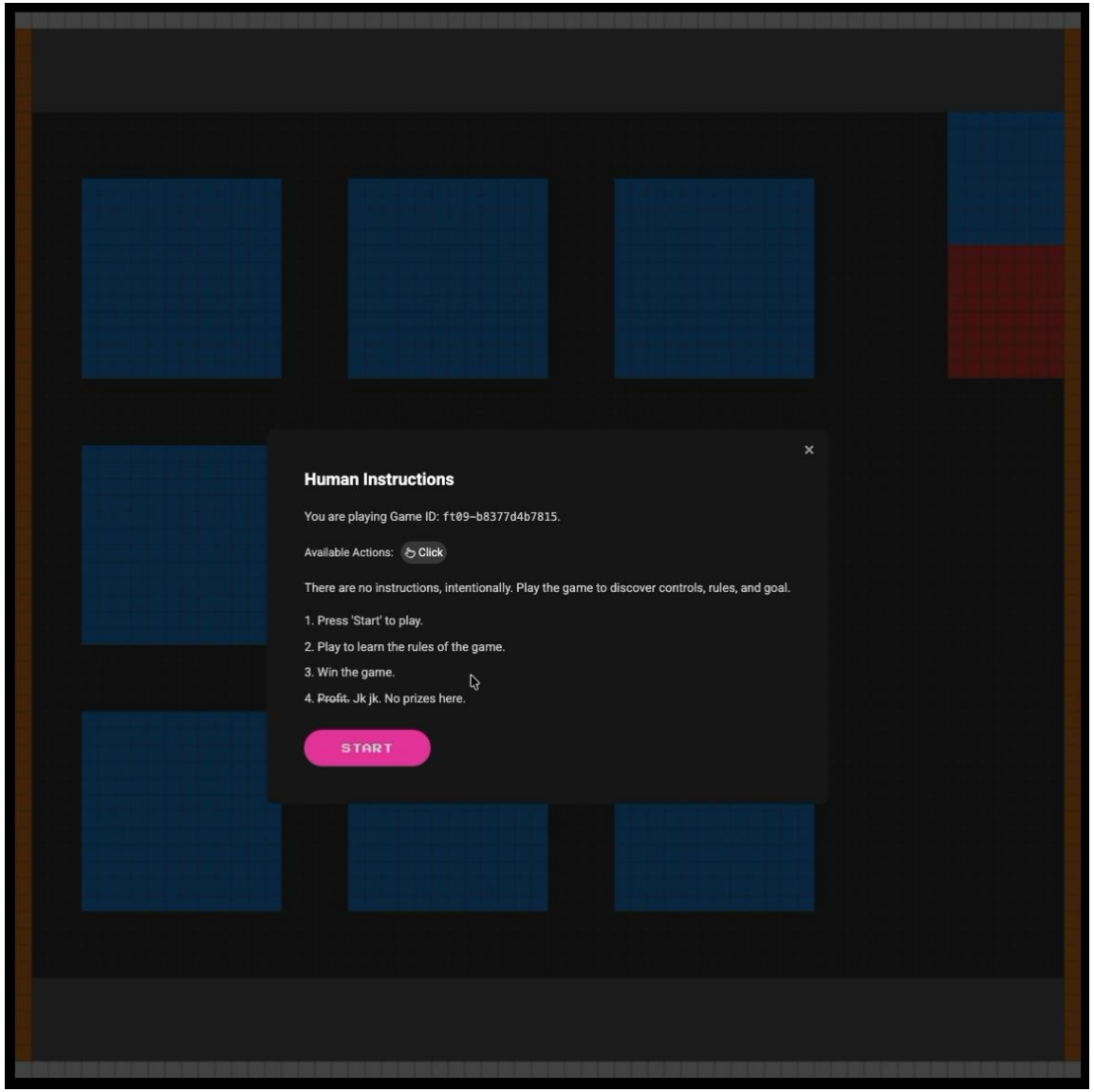
[8] [Phan et al., 1949]

ARC-AGI

- Measuring fluid intelligence
- Static puzzles
  - Somewhat comparable to IQ tests
  - Symbolic, compositional and contextual reasoning
- Interactive puzzles
  - Small games (without given rules)
  - Goal acquisition, planning, exploration

Underlying intelligence understanding:

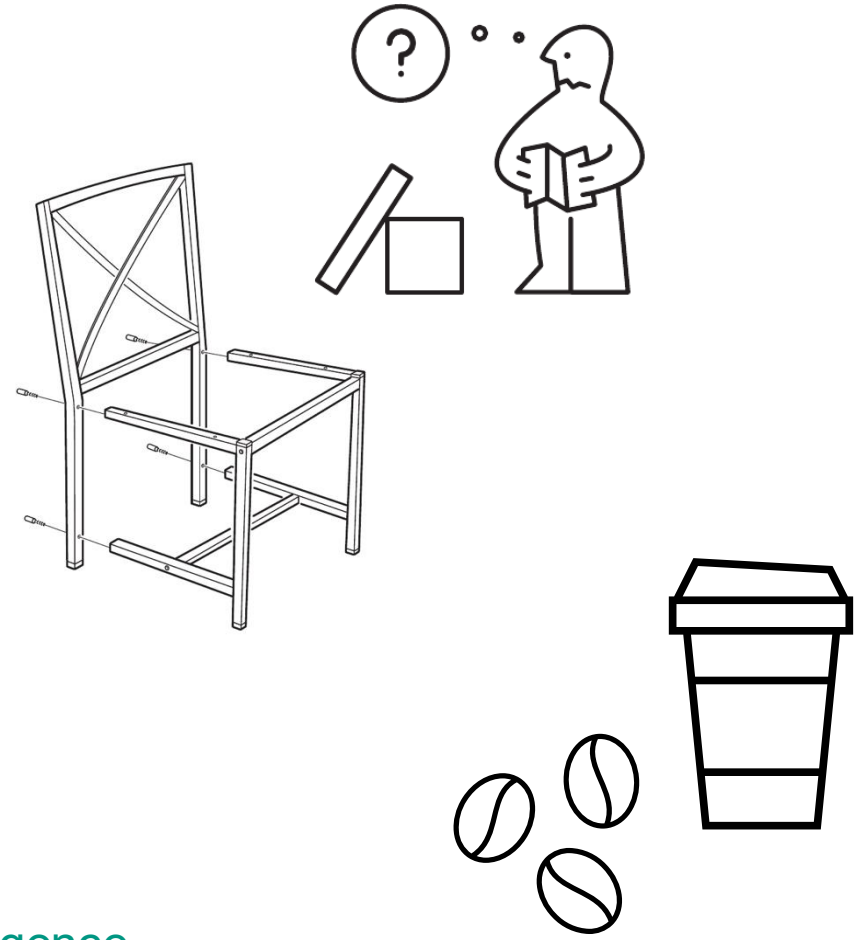
Ability to adapt, learn, and generalize



[9][Chollet et al., 2025] [10][arcprize.org]

Physical Turing Tests

- Incorporating robotics and embodied AI
- Physical goal fulfillment
  - IKEA test [11]
  - Coffee test [12]
- Physical indistinguishability
  - The Physical Turing Test [13]



Underlying intelligence understanding:

Physical presence and interaction as crucial part of intelligence

[11] [Vice, 2015] [12] [Goertzel et al., 2012] [13] [Fan, 2025] [14] [Boston Dynamics, 2021]

Benchmarking AGI

- Many different approaches
- Defined by underlying understanding of intelligence
- Specialized for certain AI subfields/modalities
  - Can be combined
  - But no guarantee for completeness
- No harmonized approach without shared definition

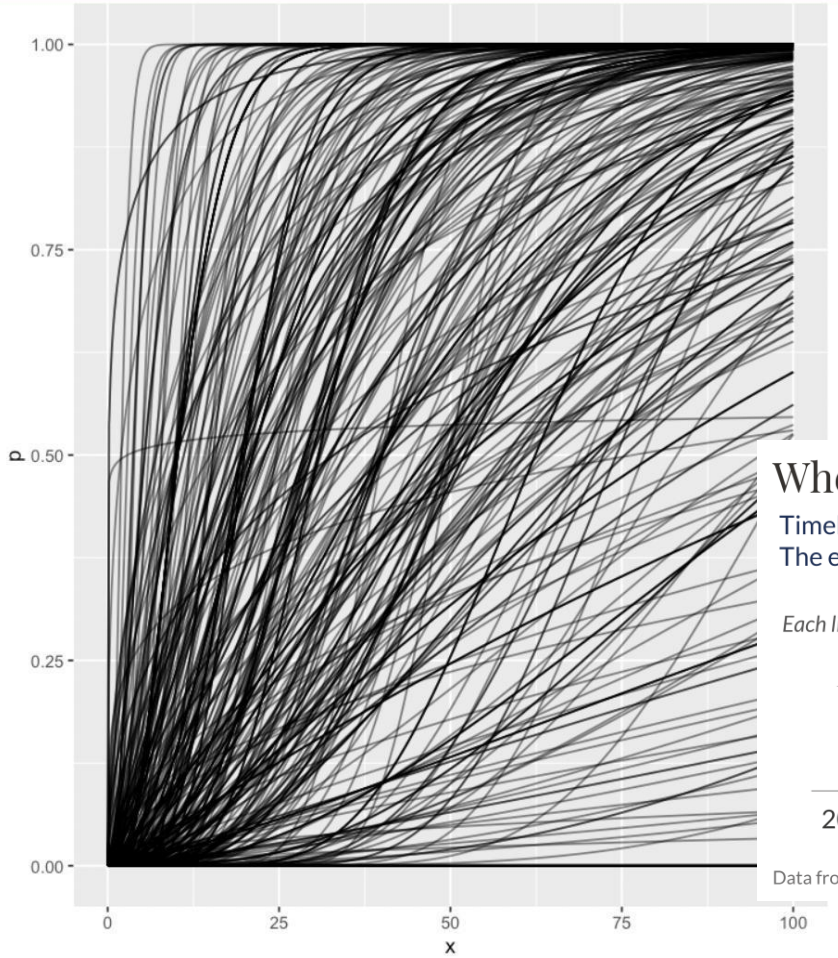
Feasibility - Timelines

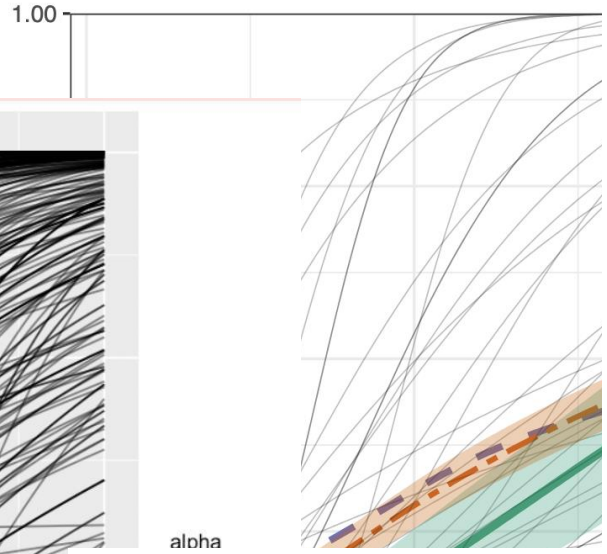
When will AGI be reached?

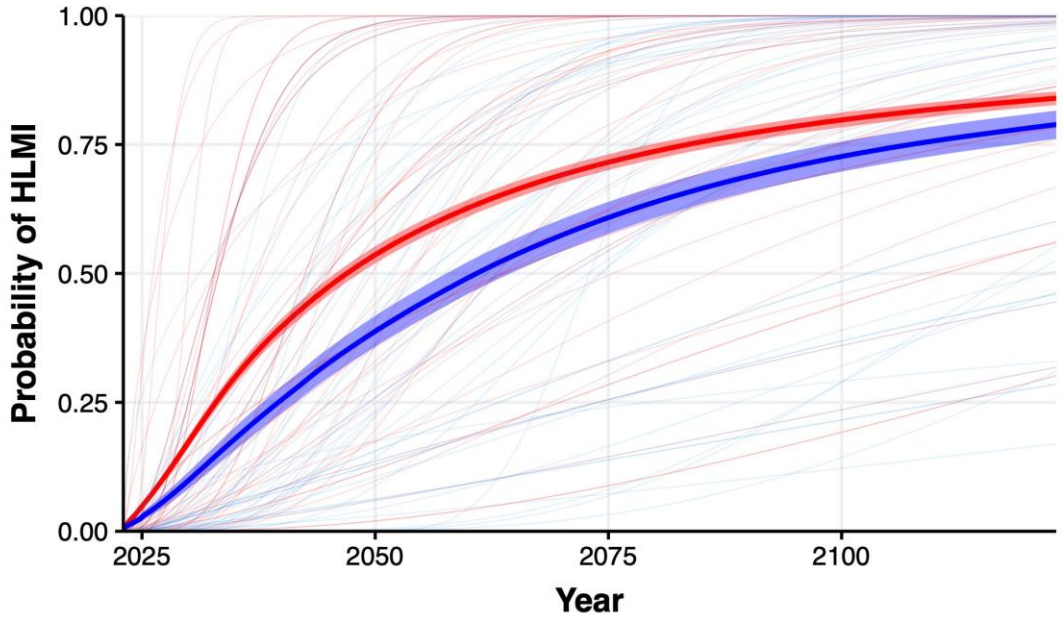


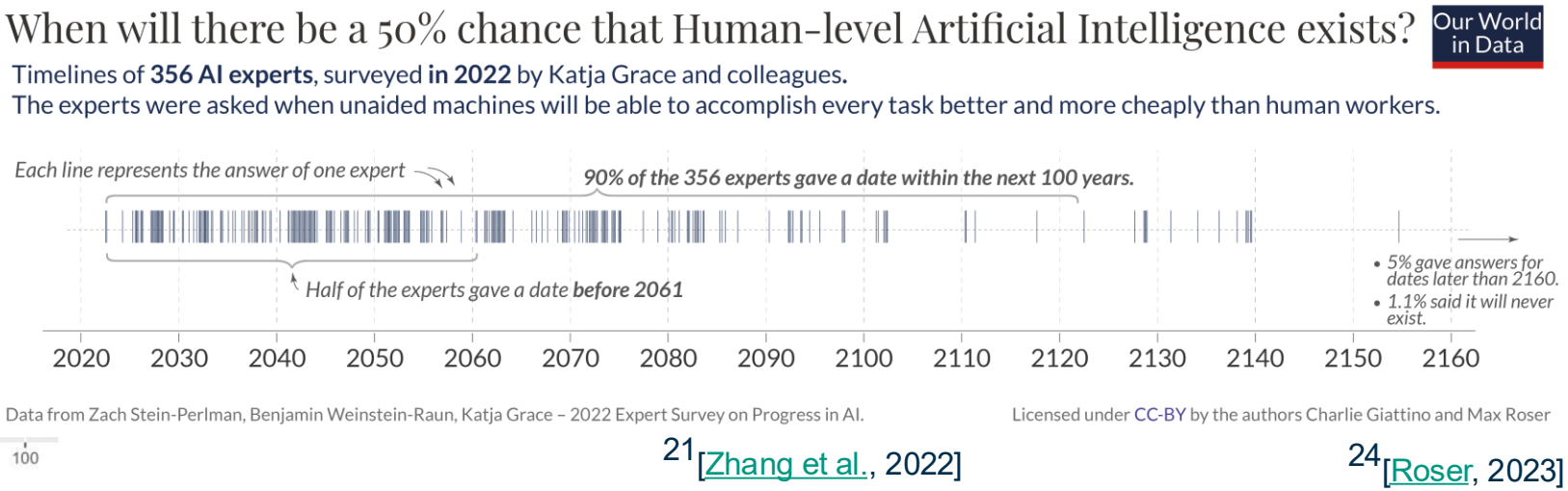
[15][Futurism, 2024] [16][Agüera y Arcas & Norvig, 2023] [17][Futurism, 2023] [18][Business Insider, 2025] [19][Dreyfus, 1992] [20][Fjelland, 2020]

Feasibility - Timelines









[23] [Grace et al., 2022]

[21] [Zhang et al., 2022]

[24] [Roser, 2023]

Feasibility - Timelines

- AGI definitions differ
- “Experts” disagree
- Predictions change

Feasibility - Approaches

1. Advancing SOTA neural networks
2. Biologically inspired AI

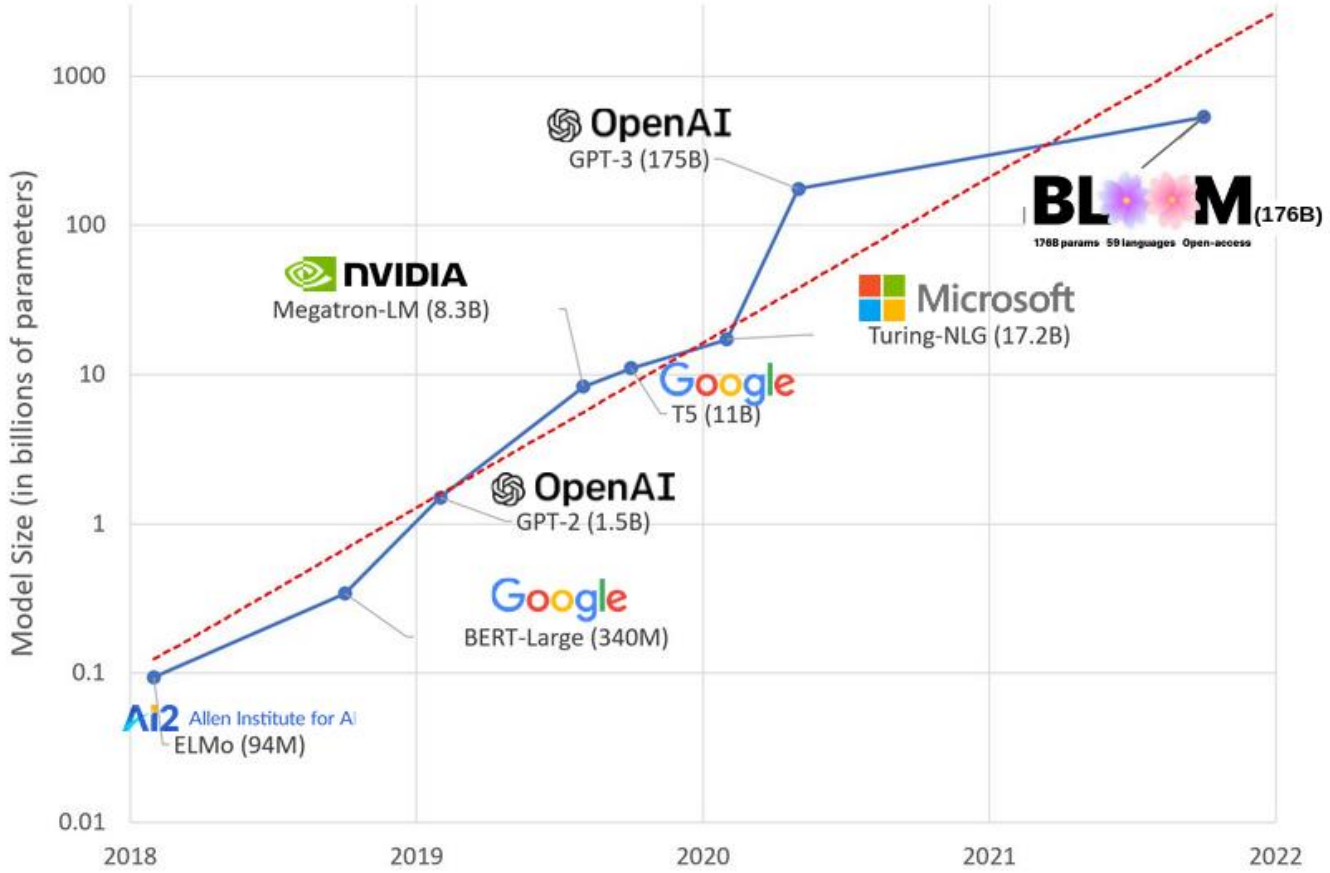
Feasibility - Better NNs

The approach:

- Larger networks
  - More complex architectures
- Increasing modalities[25]
- New learning techniques[26, 27]

The problem:

- Limited training data[28]
- Low data efficiency and generalizability[29]
- No biologic foundation[30]



[31] [ars technica, 2023]

[25] [Fei et al., 2022] [26] [Baydin et al., 2022] [27] [Ouyang, 2022] [28] [World Economic Forum, 2025] [29] [Huber et al., 2024] [30] [Sarfraz et al., 2025]

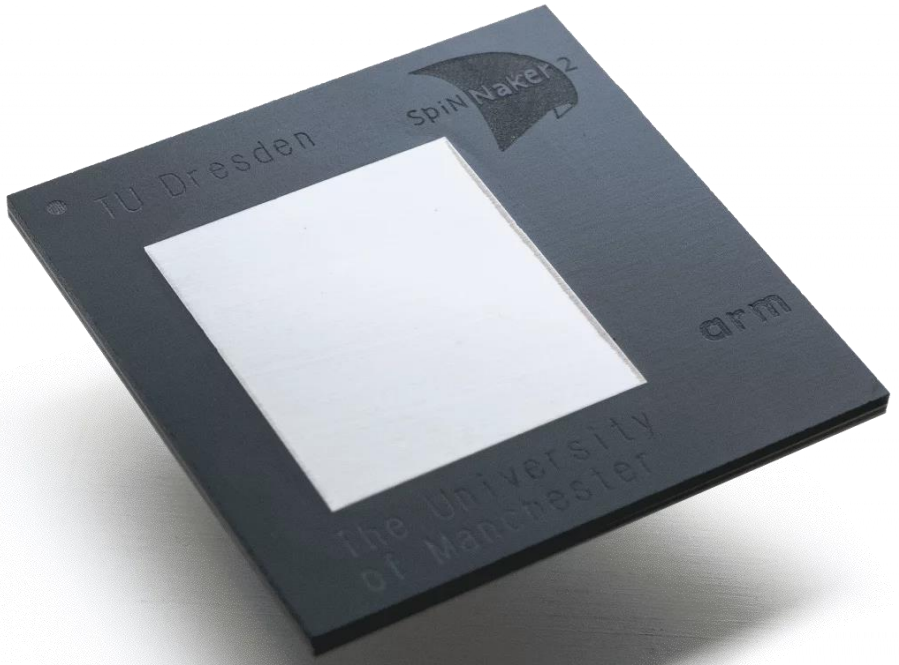
Feasibility - Bioinspiration

The approach:

- Neuromorphic computing
  - Spiking network architectures[32]
  - Specialized hardware[33]
- (Whole) brain emulation

The problem:

- Limited emulation power[34]
- Scanning resolution[34]
- Conceptual correctness[1]



[32][Maass, 2022] [33][Furber et al., 2022] [34][Zanichelli et al., 2022] [1][Russel & Norvig, 2020]

Implications

Governance

Science

Employment

Ethics

Human Creativity

Privacy

“AGI would have a big impact on several aspects of society”

Economy

Security

Social Interaction

Education

Legal Systems

Implications - Prediction Accuracy

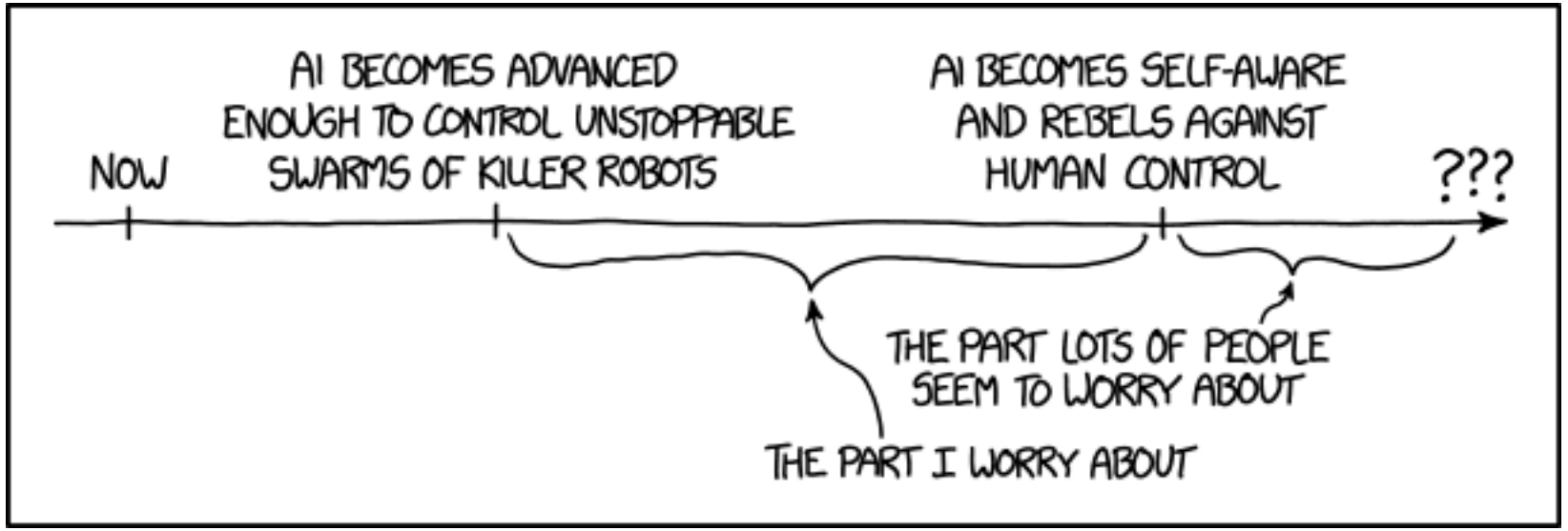
- Predictions are hard
- Increasing layers of assumptions
  - What is intelligence?
  - When is AGI achieved?
  - How does it function?
  - What properties does it have?
- Spectrum of predictions
  - Few assumptions -> broad, obvious, boring, likely correct
  - Many assumptions -> specific, interesting, likely incorrect

When encountering predictions ask yourself:

How many and which assumptions are made?

Summary

- Intelligence is not well understood
- Ideas and concepts of AGI are diverse
- This kind of debate is neither new nor useless
- Important to focus on current AI developments



[xkcd 1968]

Discussion

- Which properties would you propose for an AGI benchmark?
- Do you consider any current technology a good candidate for a development into AGI?
- What are your thoughts on surveying "experts" on AGI development timelines and feasibility?
- Compared to other societal changes how do you evaluate human-equivalent AGI?

References

0[Life Magazine, 1970]

1[Russel & Norvig, 2020]

2[Hendrycks et al., 2025]

3[Hutson, 2025]

4[Turing, 1949]

5[Weizenbaum, 1966]

6[Fedorenko et al., 2025]

7[The Verge, 2025]

8[Phan et al., 1949]

9[Chollet et al., 2025]

10[arcprize.org]

11[Vice, 2015]

12[Goertzel et al., 2012]

13[Fan, 2025]

14[Boston Dynamics, 2021]

15[Futurism, 2024]

16[Agüera y Arcas & Norvig, 2023]

17[Futurism, 2023]

18[Business Insider, 2025]

19[Dreyfus, 1992]

20[Fjelland, 2020]

21[Zhang et al., 2022]

22[Grace et al., 2023]

23[Grace et al., 2022]

24[Roser, 2023]

25[Fei et al., 2022]

26[Baydin et al., 2022]

27[Ouyang, 2022]

28[World Economic Forum, 2025]

29[Huber et al., 2024]

30[Sarfraz et al., 2025]

31[ars technica, 2023]

32[Maass, 2022]

33[Furber et al., 2022]

34[Zanichelli et al., 2022]

